1) Write a program to display the following pattern.

1
2 2
3 3 3

Ans ->

```
#include <stdio.h>

int main()
{
    int i, j;

    for (i = 1; i <= 3; i++)
    {
        for (j = 1; j <= i; j++)
        {
            printf("%d ", i);
        }

        printf("\n");
    }

    return 0;
}
```

0/p ->

```
 1
2 2
3 3 3
```

2) Write a 'C' program to display 'n' terms of the Fibonacci series using recursion.

Ans ->

```
#include <stdio.h>

int fibonacci(int n)
{
    if (n == 0)
        return 0;
    else if (n == 1)
        return 1;
    else
        return fibonacci(n - 1) + fibonacci(n - 2);
}

int main()
{
    int n, i;
```

```
    printf("Enter number of terms: ");
    scanf("%d", &n);

    printf("Fibonacci Series: ");

    for (i = 0; i < n; i++)
    {
        printf("%d ", fibonacci(i));
    }

    return 0;
}
```

O/p -> Enter number of terms: 7
Fibonacci Series: 0 1 1 2 3 5 8

3) Write a 'C' program to accept the array element from the user and display upper triangular sum of the matrix using the function.

Ans ->

```
#include <stdio.h>

void upperSum(int a[3][3])
{
    int i, j, sum = 0;

    for (i = 0; i < 3; i++)
    {
        for (j = i; j < 3; j++)
        {
            sum = sum + a[i][j];
        }
    }

    printf("Upper Triangular Sum = %d", sum);
}

int main()
{
    int a[3][3], i, j;

    printf("Enter 9 elements:\n");

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
```

```
            scanf("%d", &a[i][j]);
        }
    }

    upperSum(a);

    return 0;
}
```

Input ->
1 2 3
4 5 6
7 8 9

O/p ->
1 2 3
5 6
9